Adviser Review Memo

Semiparametric Estimation of Optimal Reserve Prices with Generative Deep Learning

Shunyu Hao October 15, 2026

Decision requested

I would like guidance on whether the current methodological and Monte Carlo paper is ready for external working-paper circulation. The decision is about the clarity and credibility of the contribution, not journal choice or another open-ended round of estimator tuning.

Paper in one sentence

When only auction maxima are observed, a structural WGAN can learn a coherent global latent value distribution, but valid reserve-price inference requires separate control of local CDF and density moments; cross-fitting and orthogonality do not compensate for missing local identifying information.

What the paper contributes

- It embeds the auction order-statistic observation rule inside a bidder-level WGAN-GP, preserving symmetric independent private values before taking the observed maximum.
- It derives a cross-fitted Neyman-orthogonal score and an asymptotic linearity result for a fixed-bandwidth regularized reserve.
- It proves the required local nuisance rate for a training-fold empirical kernel learner under primitive fixed-bandwidth and local-overlap conditions.
- Nested Monte Carlo ablations isolate the failure of WGAN-based inference in local tail moments and motivate a hybrid global-WGAN/local-score estimator.
- External-validity designs identify a local-information boundary: the formal procedure is stable with three or five bidders but deteriorates with ten bidders when maxima rarely occur near the reserve.

Claim boundary

Formal claim: valid inference for the fixed-bandwidth regularized reserve when the stated nuisance-rate, overlap, and root conditions hold; these conditions are verified for the empirical-local learner.

Diagnostic evidence: WGAN-GP global fit does not imply the local rate; Richardson correction, residual-bias envelopes, and root-union intervals for the exact optimal reserve are finite-sample prototypes, not a uniform coverage theorem.

Key evidence

- Replacing WGAN score nuisances with training-fold empirical-local moments restores near-oracle RMSE and coverage in the baseline design.
- Fixed-bandwidth target coverage is 92.0–96.5 percent across baseline, heavy-tail, and mixture distributions with three or five bidders.
- With ten bidders and weak local support, coverage falls to roughly 78 percent and multiple roots emerge.
- The stable $n^{-0.2}$ exact-target prototype reaches 96.5 percent coverage after a residual-bias envelope, but remains explicitly diagnostic.

Assumptions that should remain visible

Symmetric independent private values; observed auction maxima; known bidder count in the core design; local overlap around the reserve; and regular score root/Jacobian behavior. The paper should not imply that orthogonality repairs violations of these identifying conditions.

Three questions for the review

1. Is the local-information boundary plus the hybrid estimator a clear enough econometric contribution, or does the paper still read mainly as a WGAN application?
2. Are the formal fixed-bandwidth result and the diagnostic exact-reserve exercises separated clearly enough for the paper to circulate in its current form?
3. Which maintained auction assumption is most likely to concern a referee: symmetry/IPV, known bidder count, maxima-only observation, or local overlap?

Suggested reading order

Abstract and Introduction; Estimation and Inference; the local-nuisance and external-validity simulation results; Conclusion; then the proof appendix as needed. The identified and anonymous manuscripts, replication guide, and archived main-method results are maintained in the project repository. The two-threshold probability experiment is archived separately and does not replace the manuscript's fixed-calibration procedure.